

CS251A - Advanced Computer Architecture

HW3 - LRU Approximations

Summary

This experiment evaluates the impact of replacement policies and cache associativity on system performance, specifically analyzing IPC (Instructions Per Cycle), L1 cache Miss Rate, and L2 cache Miss Rate. The goal was to determine which policy optimizes performance and cache efficiency and whether increasing associativity significantly improves performance.

System Specification

CPU: DerivO3CPU
L1D Cache Size: 64 kB
L1I Cache Size: 32 kB
L2 Cache Size: 2 MB
CPU Frequency: 2 GHz

Associativities Tested: 8-way, 16-way

Replacement Policies Tested: LRU, NMRU, NRU, PLRU, BRRIP

- LRU (Least Recently Used)
- NMRU (Not Most Recently Used)
- NRU (Not Recently Used)
- PLRU (Pseudo-LRU)
- BRRIP (Bimodal RRIP, a thrash-resistant policy)

Workloads tested from SPEC 2017 benchmark:

- lbm (Fluid Dynamics)
- leela (AI: Monte Carlo tree search (Go))
- mcf (Route Planning)
- nab (Molecular Dynamics)
- x264 (Video compression)

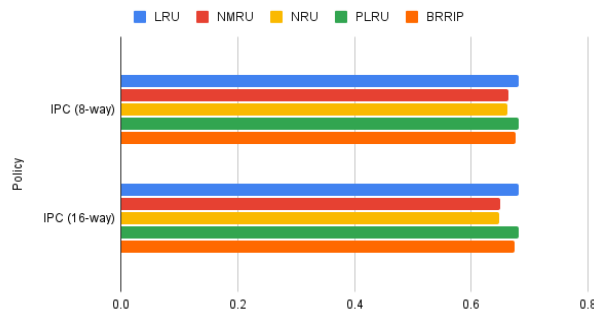
Metrics Noted:

Instructions per Cycle (IPC)
L1 Data Cache Overall Miss Rate
L2 Cache Overall Miss Rate

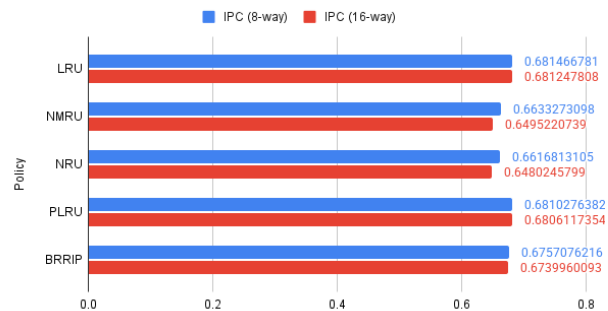
Key Findings

Impact on Performance

IPC vs Policy



IPC vs Associativity

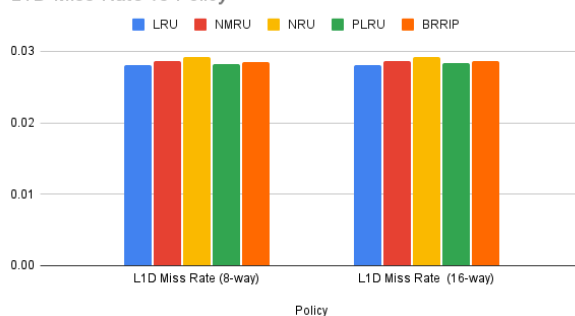


- LRU has the highest IPC, making it the best-performing policy overall.
- Pseudo-LRU policies (NMRU, NRU, PLRU) have slightly lower IPC, which suggests they do not fully approximate LRU.
- BRRIP performed almost as well as LRU, suggesting that its thrash-resistant mechanism does not harm performance.
- Increasing associativity (8-way → 16-way) has little impact on IPC, and in some cases (NMRU, NRU), it even slightly reduces IPC.

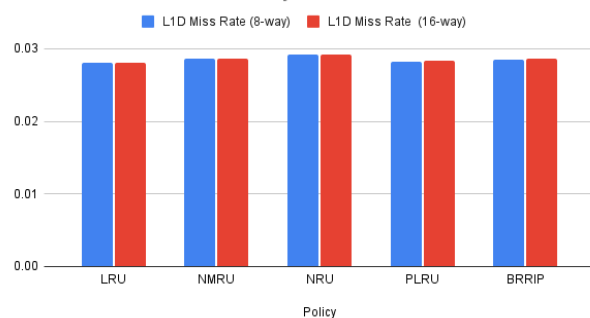
Conclusion: LRU provides the best performance, while BRRIP is a strong alternative. Increasing associativity does not significantly improve IPC.

Impact on L1 Miss Rate

L1D Miss Rate vs Policy



L1D Miss Rate vs Associativity



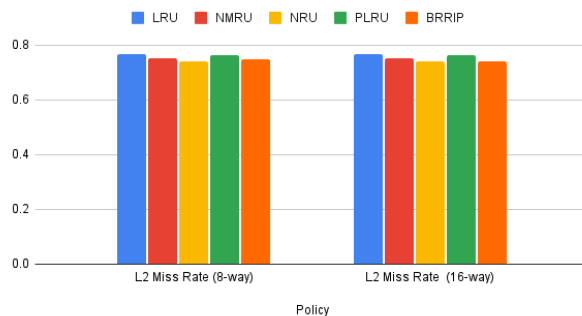
- L1 miss rates are similar across all policies, with only small variations.
- LRU has the lowest miss rate, making it effective in caching frequently accessed data.
- Pseudo-LRU policies (NMRU, NRU, PLRU) have slightly higher L1 miss rates, while BRRIP has a comparable L1 miss rate to LRU.

- Increasing associativity had negligible effect on L1 miss rate, as L1 is already optimized for recent accesses.

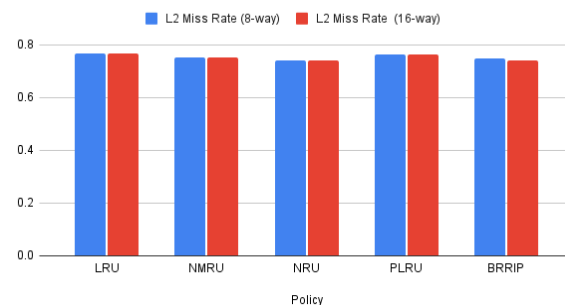
Conclusion: L1 performance is primarily dictated by replacement policy rather than associativity. LRU remains the best, while BRRIP ranks next.

Impact on L2 Miss Rate

L2 Miss Rate vs Policy



L2 Miss Rate vs Associativity



- L2 miss rates are significantly higher (~70-77%), indicating frequent misses beyond L2.
- BRRIP has the lowest L2 miss rate, confirming its effectiveness in reducing unnecessary evictions. Pseudo-LRU policies have higher L2 miss rates, reinforcing their inefficiency in long-term access patterns.
- LRU has the highest L2 miss rate, indicating that its aggressive eviction strategy sometimes removes useful data.
- Increasing associativity had a minor effect on L2 miss rates, showing that policy is more important than associativity.

Conclusion: BRRIP is the best choice for reducing L2 misses, making it useful for workloads with cache thrashing. LRU, despite being good for IPC, leads to more L2 misses.

Key Questions:

How well do the three pseudo-LRU policies approximate LRU? Do they ever do better by accident?

Pseudo-LRU policies (NMRU, NRU, PLRU) approximate LRU reasonably well but with slight inefficiencies. They never outperform LRU in IPC, and they increase miss rates slightly.

They occasionally perform well in specific workloads, but not consistently. For instance, for x264 NRU outperforms LRU for 8-way set associative cache configuration. But this seems like a one-time workload-dependent outlier.

Is the thrash-resistant policy ever useful? Can you explain when/why?

Yes, BRRIP is useful for workloads with cache thrashing. It reduces L2 miss rates significantly while maintaining IPC close to LRU. BRRIP is effective when cache reuse is unpredictable, as it prevents frequent evictions of useful cache lines.

What matters more, more associativity or a more sophisticated policy?

A more sophisticated replacement policy is more important than increasing associativity. Increasing associativity had only minor effects on performance and cache efficiency. The right replacement policy has a much greater impact on L2 efficiency and overall performance.

What's your overall recommendation? Performance is most important, and storage cost is a secondary consideration.

For best performance, LRU is the top choice, as it provides the highest IPC and lowest L1 cache miss rate. However, thrash-resistant BRRIP is the better option for workloads with cache thrashing, as it significantly reduces L2 misses while maintaining near-LRU performance.

I would not recommend Pseudo-LRU policies (NMRU, NRU, PLRU), as they offer no clear advantage. Increasing associativity (16-way) has minimal impact, so choosing the right replacement policy is far more important than increasing storage complexity.